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# A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition

JAEMYUNG KIM<sup>1</sup>, JIN-KU KANG<sup>1</sup>, (Senior Member, IEEE),  
AND YONGWOO KIM<sup>2</sup>, (Member, IEEE)

<sup>1</sup>Department of Electrical and Computer Engineering, Inha University, Incheon 22212, South Korea

<sup>2</sup>Department of System Semiconductor Engineering, Sangmyung University, Cheonan 31066, South Korea

Corresponding authors: Yongwoo Kim (yongwoo.kim@smu.ac.kr) and Jin-Ku Kang (jkang@inha.ac.kr)

**ABSTRACT** Recently, many researches have been conducted on recognition of facial emotion using convolutional neural networks (CNNs), which show excellent performance in computer vision. To obtain a high classification accuracy, a CNN architecture with many parameters and high computational complexity is required. However, this is not suitable for embedded systems where hardware resources are limited. In this paper, we present a lightweight CNN architecture optimized for embedded systems. The proposed CNN architecture has a small memory footprint and low computational complexity. Furthermore, a novel hardware-friendly quantization method that uses only integer-arithmetic is proposed. The proposed hardware-friendly quantization method maps the scale factors to power-of-two terms and replaces multiplication and division operations using scale factors with shift operations. To improve the generalization and classification performance of the CNN, we create the FERPlus-A dataset. This is a new training dataset created using a variety of image processing algorithms. After training with FERPlus-A, quantization has been performed. The size of a quantized CNN parameter is about 0.39 MB, and the number of operations is about 28 M integer operations (IOPs). By evaluating the performance of the quantized CNN that uses only integer-arithmetic on the FERPlus test dataset, the classification accuracy is approximately 86.58%. It achieved higher accuracy than other lightweight CNNs in prior studies. The proposed CNN architecture that uses only integer-arithmetic is implemented on the Xilinx ZC706 SoC platform for real-time facial emotion recognition by applying parallelism strategies and efficient data caching strategies. The FPGA-based CNN accelerator implemented for real-time facial emotion recognition achieves about 10 frame per second (FPS) at 250 MHz and consumes 2.3 W.

**INDEX TERMS** Emotion recognition, convolutional neural network, quantization, accelerator, FPGA.

## I. INTRODUCTION

Today, computers play a central role in industry and society, and are rapidly becoming a part of everyday life. Accordingly, the need for research on the interaction between humans and computers is increasing. For smooth interaction between them, a computer must be able to analyze human intention and respond accordingly. Emotions that appear in facial expressions are a universal and effective way of expressing human intentions. Analyzing human intentions through the emotions revealed in faces is called facial emotion

recognition technology, and is used in diverse fields such as automobiles and robot industries. To understand accurately the emotions displayed on human faces, a computer must recognize the face and automatically classify the emotions according to specific emotion groups.

Ekman *et al.* [1] defined seven basic emotions: happiness, anger, fear, surprise, disgust, sadness, and neutral. Moreover, they found that basic emotions are perceived in the same way regardless of human culture. In the field of computer vision, the Facial Action Coding System (FACS), a facial expression analysis method proposed in [2], was used as a typical facial emotion recognition model. FACS defines the basic movement of the facial muscles as Action

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